

AMATH 353 HW 6

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Question 1: Method of Characteristics (part 1)

(a) Find the characteristic curves of the PDE

$$u_t + xu_x = 0.$$

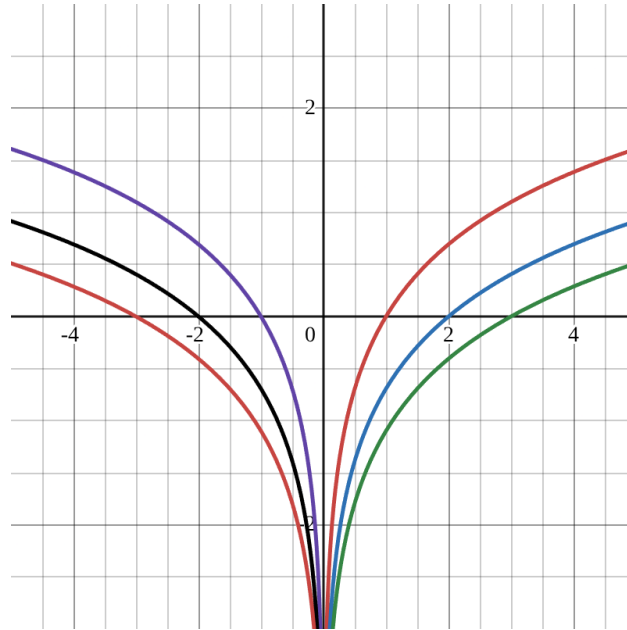
Sketch/plot the characteristic curves in the xt -plane. What do you predict the dynamics to look like?

$$\begin{aligned}u_t + xu_x &= 0 \\ \frac{d}{dt}u(x(t), t) &= u_t + \frac{dx}{dt}u_x \\ u_t + \frac{dx}{dt}u_x &= u_t + xu_x \\ \frac{dx}{dt} &= x\end{aligned}$$

Solve the ODE:

$$\begin{aligned}\frac{1}{x}dx &= dt \\ \int \frac{1}{x}dx &= \int dt \\ \ln|x| &= t + \ln|x_0| \\ x &= e^{t+\ln|x_0|} \\ x &= x_0e^t\end{aligned}$$

The wave will go faster the further it gets from the origin. If $x_0 > 0$ it moves to the right. If $x_0 < 0$ it moves to the left.



(b) Use the method of characteristics to solve

$$\begin{cases} u_t + tu_x = 0, & x \in \mathbb{R}, t \geq 0, \\ u(x, 0) = \cos(x), & x \in \mathbb{R}, t \geq 0. \end{cases}$$

Give a qualitative description of the dynamics of the solution.

$$\begin{aligned} u_t + tu_x &= 0 \\ \frac{d}{dt}u((x), t) &= u_t + \frac{dx}{dt}u_x \\ u_t + tu_x &= u_t + \frac{dx}{dt}u_x \\ \frac{dx}{dt} &= t \end{aligned}$$

Solve the ODE

$$\begin{aligned}\frac{dx}{dt} &= t \\ dx &= t dt \\ \int dx &= \int t dt \\ x &= \frac{t^2}{2} + x_0 \\ x_0 &= x - \frac{t^2}{2} \\ u(x_0, 0) &= \cos\left(x - \frac{t^2}{2}\right)\end{aligned}$$

$$u(x, t) = \cos\left(x - \frac{t^2}{2}\right)$$

The wave speeds up as it moves away from x_0 , and always moves to the right.

(c) Use the method of characteristics to solve

$$\begin{cases} u_t + \cos(t)u_x = 0, & x \in \mathbb{R}, t \geq 0, \\ u(x, 0) = e^{-x^2}, & x \in \mathbb{R}, t \geq 0. \end{cases}$$

Give a qualitative description of the dynamics of the solution.

$$\begin{aligned}u_t &= \cos(t)u_x = 0 \\ \frac{d}{dt}u(x(t), t) &= u_t + \frac{dx}{dt}u_x \\ u_t + \frac{dx}{dt}u_x &= u_t + \cos(t)u_x \\ \frac{dx}{dt} &= \cos(t)\end{aligned}$$

Solve the ODE

$$\begin{aligned}\frac{dx}{dt} &= \cos(t) \\ dx &= \cos(t)dt \\ \int dx &= \int \cos(t)dt \\ x &= \sin(t) + x_0 \\ x_0 &= x - \sin(t)\end{aligned}$$

$$\begin{aligned}u(x_0, 0) &= e^{-(x - \sin(t))^2} \\ u(x, t) &= e^{-(x - \sin(t))^2}\end{aligned}$$

The wave bounces between -1 and 1 , and the velocity depends on time.

Question 2: Method of Characteristics (part 2)

(a) Determine the (implicit) solution to the nonlinear, homogeneous PDE problem

$$\begin{cases} u_t + u^2 u_x = 0, & x \in \mathbb{R}, t \geq 0, \\ u(x, 0) = \tanh(x), & x \in \mathbb{R}. \end{cases}$$

$$\begin{aligned} x &= x(t) \\ \frac{d}{dt}u(x(t), t) &= u_t + \frac{dx}{dt}u_x \\ u_t + \frac{dx}{dt}u_x &= u_t = u^2 u_x \\ \frac{dx}{dt} &= u^2 \end{aligned}$$

Solve The ODE

$$\begin{aligned} \frac{dx}{dt} &= u^2 \\ dx &= u^2 dt \\ \int dx &= \int u^2 dt \\ x &= u^2 t + x_0 \\ x_0 &= x - u^2 t \\ u(x_0, 0) &= \tanh(x_0) \\ u(x, t) &= \tanh(x_0) \\ u(x, t) &= \tanh(x - u^2 t) \end{aligned}$$

(b) Determine the (explicit) solution to the linear, in-homogeneous PDE problem

$$\begin{cases} u_t + 2tu_x = 2t \cos(x), & x \in \mathbb{R}, t \geq 0, \\ u(x, 0) = \operatorname{sech}^2(x), & x \in \mathbb{R}. \end{cases}$$

$$\begin{aligned} \frac{d}{dt}u(x(t), t) &= u_t + \frac{dx}{dt}u_x \\ u_t + \frac{dx}{dt}u_x &= u_t + 2tu_x \\ \frac{dx}{dt} &= 2t \end{aligned}$$

Solve the ODE

$$\begin{aligned}\frac{dx}{dt} &= 2t \\ dx &= 2t dt \\ \int dx &= \int 2t dt \\ x &= t^2 + x_0\end{aligned}$$

$$\begin{aligned}\frac{du}{dt} &= 2t \cos(x(t)) \\ \int du &= \int 2t \cos(t^2 + x_0) dt \\ u(t) &= \sin(t^2 + x_0) + C\end{aligned}$$

Initial Conditions

$$\begin{aligned}t = 0, \quad u(x, 0) &= \operatorname{sech}^2(x) \\ t = 0, \quad x(0) &= x_0 \\ t = 0, \quad u(0) &= \sin(x_0) + C \\ \sin(x_0) + C &= \operatorname{sech}^2(x_0) \\ C &= \operatorname{sech}^2(x_0) - \sin(x_0)\end{aligned}$$

$$u(x_0, t) = \sin(t^2 + x_0) + \operatorname{sech}^2(x_0) - \sin(x_0)$$

$$\begin{aligned}x &= t^2 + x_0 \\ x_0 &= x - t^2 \\ u(x, t) &= \sin(t^2 + (x - t^2)) + \operatorname{sech}^2(x - t^2) - \sin(x - t^2)\end{aligned}$$

(c) Determine the (explicit) solution to the non-linear, in-homogeneous PDE problem

$$\begin{cases} u_t + uu_x = \cos(t), & x \in \mathbb{R}, t \geq 0, \\ u(x, 0) = x, & x \in \mathbb{R}. \end{cases}$$

$$\begin{aligned}\frac{d}{dt}u(x(t), t) &= u_t + \frac{dx}{dt}u_x \\ u_t + \frac{dx}{dt}u_x &= u_t + uu_x \\ \frac{dx}{dt} &= u\end{aligned}$$

Solve for u

$$\begin{aligned}\frac{du}{dt} &= \cos(t) \\ \int du &= \int \cos(t) dt \\ u(t) &= \sin(t) + C\end{aligned}$$

Initial Conditions:

$$\begin{aligned}t = 0, \quad u(0) &= \sin(0) + C = C \\ C &= x_0 \\ u(t) &= \sin(t) + x_0\end{aligned}$$

Solve for x

$$\begin{aligned}\frac{dx}{dt} &= u = \sin(t) + x_0 \\ \int dx &= \int (\sin(t) + x_0) dt \\ x(t) &= -\cos(t) + x_0 t + C_2 \\ t = 0, \quad x(0) &= -\cos(0) + x_0(0) + C_2 = -1 + C_2 = x_0 \\ C_2 &= x_0 + 1 \\ x(t) &= -\cos(t) + x_0 t + x_0 + 1 \\ x_0(t+1) &= x + \cos(t) - 1 \\ x_0 &= \frac{x + \cos(t) - 1}{t + 1} \\ u(x, t) &= \sin(t) + \frac{x + \cos(t) - 1}{t + 1}\end{aligned}$$

Question 3: Breaking Time

The breaking time t_b is the first (non-negative) time at which characteristics intersect, and we computed

$$t_b = \min_{x_0 \in \mathbb{R}} \left\{ \frac{-1}{c'(u_0(x_0))u'_0(x_0)} \geq 0 \right\} \quad \text{for} \quad \begin{cases} u_t + c(u)u_x = 0, & x \in \mathbb{R}, t \geq 0, \\ u(x, 0) = u_0(x), & x \in \mathbb{R}. \end{cases}$$

For each of the following problems, determine whether the solutions breaks and if so, compute the breaking point (x_b, t_b) .

- (a) The non-viscous traffic model $u_t + v_1(1 - \frac{u}{u_1})u_x = 0$ with initial profile $u_0(x) = \frac{u_1}{2}e^{-(x/\beta)^2}$. How does β impact the initial profile and the breaking time?

$$\begin{aligned} c(u) &= v_1 \left(1 - \frac{u}{u_1} \right) \\ c'(u) &= -\frac{v_1}{u_1} \\ u'_0(x_0) &= \frac{u_1}{2} e^{-(x_0/\beta)^2} \left(\frac{-2x_0}{\beta^2} \right) = -u_1 \frac{x_0}{\beta^2} e^{-(x_0/\beta)^2} \\ t_b &= \frac{-1}{\left(-\frac{v_1}{u_1} \right) \left(-u_1 \frac{x_0}{\beta^2} e^{-(x_0/\beta)^2} \right)} \end{aligned}$$

We take the derivative of the denominator with respect to x_0 and set it to 0 to find the minimum time

$$\begin{aligned} \frac{d}{dx_0} \left(-\frac{v_1}{\beta^2} x_0 e^{-(x_0/\beta)^2} \right) &= \frac{v_1}{\beta^2} e^{-(x_0/\beta)^2} \left[1 - \frac{2x_0^2}{\beta^2} \right] = 0 \\ 1 - \frac{2x_0^2}{\beta^2} &= 0 \implies x_0 = \pm \frac{\beta}{\sqrt{2}} \end{aligned}$$

Using $x_0 = -\frac{\beta}{\sqrt{2}}$ to make the denominator positive (and t_b positive):

$$\begin{aligned} t_b &= \frac{-1}{-\frac{v_1}{\beta^2} \left(\frac{\beta}{\sqrt{2}} \right) e^{-1/2}} = \frac{\beta\sqrt{2}e}{v_1} \\ x(t) &= c(u_0)t + x_0 \\ u_0 \left(-\frac{\beta}{\sqrt{2}} \right) &= \frac{u_1}{2} e^{-1/2} = \frac{u_1}{2\sqrt{e}} \\ c(u_0) &= v_1 \left(1 - \frac{1}{2\sqrt{e}} \right) \\ x_b &= v_1 \left(1 - \frac{1}{2\sqrt{e}} \right) \left(\frac{\beta\sqrt{2}e}{v_1} \right) - \frac{\beta}{\sqrt{2}} \\ x_b &= \beta\sqrt{2}e - \frac{\beta\sqrt{2}}{2} - \frac{\beta\sqrt{2}}{2} = \beta\sqrt{2}(\sqrt{e} - 1) \end{aligned}$$

If β is small, the change is steep, so the cars that end up stopping move fast. If β is large, the cars stop more gradually.

(b) The wacky PDE $u_t + \sinh\left(\sqrt{1 + \exp(u)}\right)u_x = 0$ with initial profile $u_0(x) = \tanh(x)$.

$$\begin{aligned}c(u) &= \sinh\left(\sqrt{1 + \exp(u)}\right) \\c'(u) &= \cosh\left(\sqrt{1 + \exp(u)}\right) \frac{e^u}{2\sqrt{1 + e^u}} \\u_0(x_0) &= \tanh(x_0) \\u'_0(x_0) &= \operatorname{sech}^2(x_0)\end{aligned}$$

The solution breaks when $t_b = \frac{-1}{c'(u_0(x_0))u'_0(x_0)} > 0$. Because $c'(u)$ is always positive and $u'_0(x_0) = \operatorname{sech}^2(x_0)$ is always positive, the product $c'(u_0(x_0))u'_0(x_0)$ is always positive. Therefore, t_b would be negative, meaning the solution does not break for $t \geq 0$.

(c) Find a PDE $u_t + c(x, t)u_x = 0$ and initial profile $u_0(x)$ so that the solution becomes arbitrarily steep as $t \rightarrow +\infty$ but never breaks.

We want our characteristic curves to get close but never break. Let $c(x, t) = -x$.

$$u_t - xu_x = 0$$

Let $u_0(x) = \cos(x)$.

$$\begin{aligned}\frac{dx}{dt} &= -x \implies \int \frac{1}{x} dx = \int -1 dt \\ \ln|x| &= -t + C \implies x(t) = x_0 e^{-t} \\ x_0 &= x e^t \\ u(x, t) &= u_0(x_0) = \cos(x e^t) \\ u_x(x, t) &= -\sin(x e^t) e^t\end{aligned}$$

As $t \rightarrow \infty$, $e^t \rightarrow \infty$, making the slope u_x steep, but it never breaks.

Question 4: Understanding Qualitative Behavior

In this problem, you will (rigorously) establish several results concerning the qualitative behavior of solutions to large classes of conservation laws. Consider the PDE problem given by

$$\begin{cases} u_t + c(x, t, u)u_x = 0, & x \in \mathbb{R}, t \geq 0, \\ u(x, 0) = u_0(x), & x \in \mathbb{R}. \end{cases}$$

You may assume that $c(x, t, u)$ and $u_0(x)$ are smooth functions, i.e., you can safely differentiate them.

(a) Prove that if $c = c(t)$, then the profile never deforms, i.e., $u(x, t)$ is a shift of the initial condition.

Assume $c = c(t)$. The PDE is $u_t + c(t)u_x = 0$ with $u(x, 0) = u_0(x)$. Solve using the method of characteristics:

$$\begin{aligned} \frac{dx}{dt} &= c(t) \\ dx &= c(t)dt \implies x = \int c(t)dt + x_0 \\ x_0 &= x - \int c(t)dt \\ u(x, t) &= u(x_0, 0) = u_0(x_0) = u_0\left(x - \int c(t)dt\right) \end{aligned}$$

This shows that $u(x, t)$ is just a shift of the original initial condition $u_0(x)$ by $\int c(t)dt$. The profile never deforms.

(b) Prove that if $c = c(x) > 0$, then the solution never breaks, regardless of $u_0(x)$.

Assume $c = c(x) > 0$. The characteristics are given by:

$$\begin{aligned} \frac{dx}{dt} &= c(x) \\ \frac{du}{dt} &= 0 \implies u(x, t) = u_0(x_0) \end{aligned}$$

Since $c(x)$ is a smooth function, we can use the existence and uniqueness theorem for ODEs. The theorem says that for any point (x, t) , there is exactly one solution curve passing through it. Therefore, characteristic curves starting at different points x_1 and x_2 can never intersect, meaning the solution never breaks.

(c) Prove that if $c = c(u) \neq \text{const.}$, then there exists an initial profile $u_0(x)$ corresponding to a solution which breaks in finite time.

Assume $c = c(u)$ is not constant. By the method of characteristics:

$$\begin{aligned} x(t) &= c(u_0(x_0))t + x_0 \\ u(x, t) &= u_0(x_0) \end{aligned}$$

Taking the partial derivative with respect to x :

$$\begin{aligned} u_x &= u'_0(x_0) \frac{\partial x_0}{\partial x} \\ 1 &= c'(u_0(x_0)) u'_0(x_0) \frac{\partial x_0}{\partial x} t + \frac{\partial x_0}{\partial x} \\ \frac{\partial x_0}{\partial x} &= \frac{1}{1 + c'(u_0(x_0)) u'_0(x_0) t} \end{aligned}$$

The solution breaks when the denominator is zero. Thus, we need $c'(u_0(x_0)) u'_0(x_0) < 0$. Because c is smooth and non-constant, there must be some u where $c'(u) \neq 0$. If $c'(u) > 0$, we can choose an initial profile $u_0(x)$ with a negative slope at that point. If $c'(u) < 0$, we can choose an initial profile $u_0(x)$ with a positive slope at that point. In either case, we can make the product $c'(u_0(x_0)) u'_0(x_0)$ strictly negative, guaranteeing that the solution will break in finite time.

(d) Prove that if $c = c(u)$ and $u_0(x)$ are both increasing functions, then the solution will never break.

Assume $c = c(u)$ and $u_0(x)$ are both increasing functions. This implies that $c'(u) > 0$ and $u'_0(x) > 0$ for all valid inputs. The breaking time is given by:

$$t_b = \frac{-1}{c'(u_0(x_0)) u'_0(x_0)}$$

Since both c' and u'_0 are strictly positive, their product $c'(u_0(x_0)) u'_0(x_0)$ is also strictly positive. Therefore, t_b will be strictly negative. Since we only care about $t \geq 0$, the solution will never break.